GLA UNIVERSITY, GREATER NOIDA

Greater Noida - 201310, Uttar Pradesh, India

Title: IPL Match Outcome Predictor: Win Probability Engine for Team Strategy

SOFTWARE REQUIREMENT SPECIFICATION

Version 1.0

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING - AI

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Revision History

Version	Date	Change	Reason
1.0	04-09-2026	Initial version	First submission

1. Introduction

1.1 Purpose

This document specifies the requirements of the IPL Match Outcome Predictor: Win Probability Engine for Team Strategy, a supervised machine-learning system for pre-match IPL outcome prediction. It defines the expected data processing, modelling, evaluation, visualization, and user-interface behaviour against which implementation and testing will be verified.

1.2 Scope

The system is designed for an IPL team analyst or captain who wants a simple pre-match estimate of which of two selected teams is more likely to win. The analyst provides Team 1, Team 2, venue, season, toss winner, and toss decision. The system uses historical information available before the selected match to generate features, predicts the likely winner, and presents estimated win probabilities together with supporting team, venue, toss, and head-to-head insights.

The project uses one public IPL match-level CSV dataset. It does not require a second dataset or ball-by-ball data. The system is a historical/pre-match prediction tool, not a live in-match predictor and not a betting recommendation system.

1.3 Definitions

Term	Definition
Pre-match prediction	Prediction made using information available before the selected match begins.
Win probability	Model-estimated probability that a team will win the selected match.
Level 0 baseline	A baseline that ignores match inputs and always predicts the overall majority class.
Level 1 baseline	A simple baseline that uses one obvious feature, such as Team 1, and predicts its historical majority outcome.
Precision@1	Precision when only the single highest-probability predicted winner is selected (K=1).
Data leakage	Use of information from the selected match or future matches that would not be available before prediction.

2. Overall Description

2.1 Product Perspective

The system is a Python-based machine-learning application with a Streamlit user interface and a separate business-intelligence/dashboard component for analysis. The complete system workflow is shown in Figure 2.1.

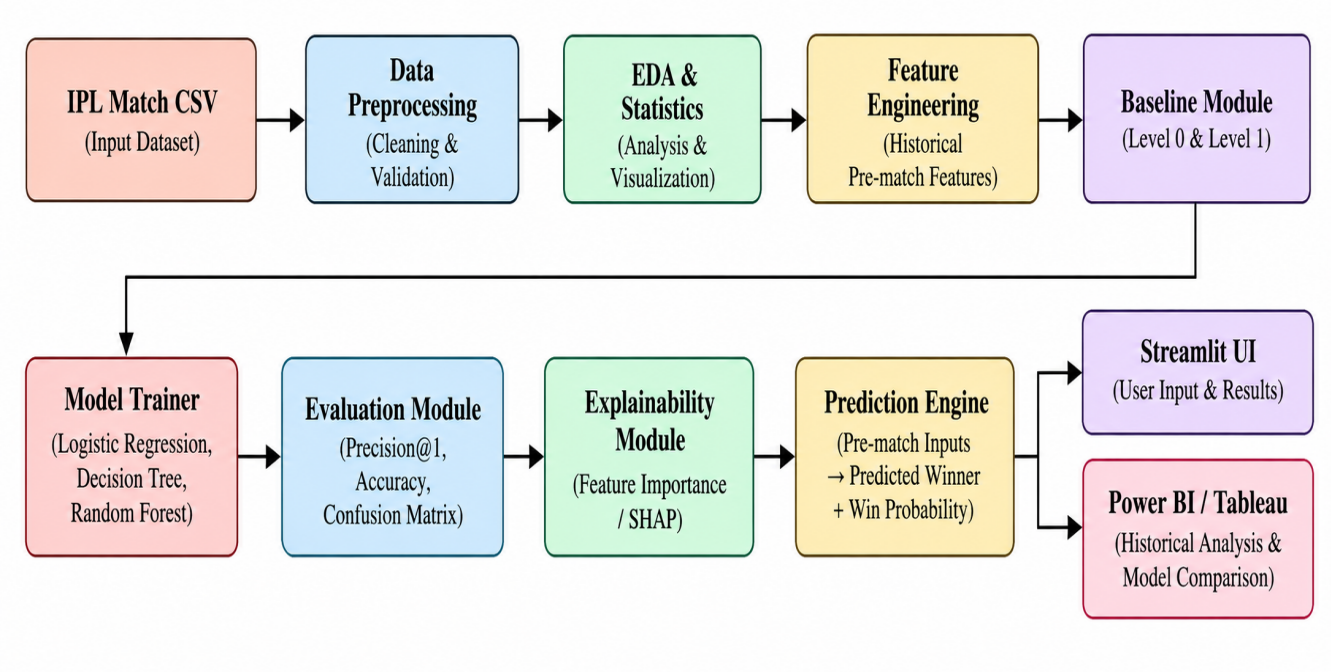


Figure 2.1 - System Workflow

The workflow shows the movement of data from the single IPL match-level CSV through preprocessing, exploratory analysis, historical feature engineering, baseline evaluation, machine-learning models, model evaluation, explainability, prediction, and final presentation through Streamlit and Power BI/Tableau.

2.2 User Classes

User Class	Who They Are	What They May Do
IPL Team Analyst / Captain	Primary decision-support user	Select pre-match conditions, view predicted winner and probabilities, inspect historical performance and model factors.
Project Evaluator / Faculty	Reviewer of the academic project	Inspect methodology, model comparison, dashboard, testing, and documentation.

2.3 Assumptions and Dependencies

• The project depends on the public Kaggle dataset: IPL All Matches 2008-2025.

• The core application depends on Python, Pandas, NumPy, scikit-learn, Matplotlib/Seaborn/Plotly, and Streamlit.

• Power BI or Tableau may be used for the analyst dashboard.

• Historical features are calculated using information available before the prediction match.

• The dataset is treated as the single data source for the project.

2.4 Constraints

• The project is limited to information contained in the selected match-level CSV.

• No real-time match feed is required.

• No live in-match prediction is included.

• No betting recommendation is included.

• No advanced deep-learning model is required.

• Performance claims will be based on measured test results rather than assumed accuracy.

3. Data Source and Baseline

Item	Value
Dataset name	IPL All Matches 2008-2025
Source and URL	https://www.kaggle.com/datasets/dhrubangtalukdar/indian-premier-league-ipl-all-matches2008-2025
Dataset format	Single CSV file
Prediction target	Whether Team 1 wins the selected match
Task type	Binary classification
Primary evaluation metric	Precision@1 (K=1)
Metric rationale	The system selects one predicted winner; Precision@1 directly evaluates whether the single selected winner is correct.
Level 0 baseline	Always predicts the overall majority class, ignoring all match inputs. Score will be measured on the defined test set.
Level 1 baseline	Uses one obvious feature, Team 1, and predicts Team 1's historical majority outcome. Score will be measured on the defined test set.
ML models	Logistic Regression, Decision Tree, Random Forest
Improvement measure	((Model Precision@1 - Level 1 Precision@1) / Level 1 Precision@1) x 100
Known data-quality checks	Missing values, duplicate rows, inconsistent categorical labels, date/season consistency, and invalid outcome records.
Release target	The selected model should be preferred only if its measured Precision@1 improves on Level 1; otherwise the result will be reported as a legitimate finding.

3.1 Data Leakage Control

The winner and other post-match outcome information shall not be used as input features. Fields such as final winner, final scores, result margin, player of the match, and any other information that becomes known only after the match shall be excluded from prediction inputs. Historical team, venue, and head-to-head statistics shall be calculated using only matches that occurred before the prediction match.

4. System Architecture

The system is organized as modular layers so that data preparation, analysis, feature engineering, modelling, evaluation, prediction, and presentation can be developed and tested independently. Each component has a defined responsibility and exchanges processed information with the next stage.

Component	Detailed Responsibility
CSV Data Layer	Stores and loads the single IPL match-level CSV dataset. Preserves the raw source unchanged and supplies match records such as teams, venue, city, season, match date, toss information and outcome fields to downstream modules.
Data Preprocessor	Validates the dataset schema and required columns, checks missing values and duplicate records, converts dates and data types, standardizes team and venue names, and removes or handles invalid records. Produces a separate cleaned dataset for analysis and modelling.
EDA & Statistics	Performs exploratory analysis of teams, venues, seasons, tosses and match outcomes. Calculates descriptive statistics, win percentages, probabilities and correlations, and generates visualizations to identify useful patterns and relationships before modelling.
Feature Engineer	Creates pre-match historical features such as team win rate, recent form, head-to-head performance, venue performance, toss information and season-related features. Features are calculated chronologically using only information available before the prediction match to prevent future-data leakage.
Baseline Module	Provides benchmark predictions before machine-learning models are trained. Level 0 ignores match inputs and predicts the overall majority class. Level 1 uses the selected team's historical majority outcome. Baseline results are evaluated on the same defined test procedure used for model comparison.
Model Trainer	Trains supervised binary-classification models using the prepared pre-match features. The planned models are Logistic Regression, Decision Tree and Random Forest. Training uses a reproducible evaluation setup suitable for chronological sports data, with preprocessing applied consistently.
Evaluation Module	Evaluates predictions on unseen test data using Precision@1 as the primary metric, with accuracy, confusion matrix and other supporting metrics where required. Compares every ML model with Level 0 and Level 1 and calculates percentage improvement over Level 1.
Prediction Engine	Acts as the application's prediction service. Accepts Team 1, Team 2, venue, season, toss winner and toss decision, applies the same feature-engineering and preprocessing logic used during training, and returns the predicted winner together with class probabilities.
Explainability Module	Provides understandable evidence for model predictions by identifying influential features. Uses model feature importance for suitable tree-based models and SHAP where appropriate, helping the analyst understand which historical or pre-match factors contributed to the prediction.
Streamlit UI	Provides the interactive user interface for the IPL team analyst or captain. Collects valid pre-match inputs, triggers the Prediction Engine, and displays the predicted winner, win probabilities, historical supporting statistics and relevant model insights.
Power BI / Tableau	Provides an analyst-facing dashboard for historical and model-comparison analysis. Presents team performance, venue trends, toss analysis, season trends, head-to-head information and baseline-versus-model evaluation results using interactive filters and visualizations.

Architecture Data Flow

IPL CSV -> Data Preprocessor -> EDA & Statistics -> Feature Engineer -> Baseline Module -> Model Trainer -> Evaluation Module -> Explainability / Prediction Engine -> Streamlit UI and Power BI / Tableau.

The architecture follows a modular design so that changes to one component can be tested without rewriting the entire system. The raw CSV remains unchanged, processed data is stored separately, and post-match information is excluded from prediction inputs.

5. Functional Requirements

5.1 Data and Preprocessing Requirements

ID	Functional Requirement
FR-1.1	The system shall load the selected IPL CSV dataset and validate that required columns are present.
FR-1.2	The system shall check for missing values and handle them according to documented preprocessing rules.
FR-1.3	The system shall identify and handle duplicate records.
FR-1.4	The system shall validate data types, dates, seasons, categorical values, and outcome records.
FR-1.5	The system shall standardize inconsistent team and venue names when required.

5.2 Feature, Model, and Evaluation Requirements

ID	Functional Requirement
FR-2.1	The system shall create pre-match historical features using only information available before the prediction match.
FR-2.2	The system shall implement Level 0 and Level 1 baselines.
FR-2.3	The system shall train Logistic Regression, Decision Tree, and Random Forest classifiers.
FR-2.4	The system shall calculate Precision@1 as the primary evaluation metric.
FR-2.5	The system shall compare ML models with Level 0 and Level 1 baselines.
FR-2.6	The system shall calculate percentage improvement over the Level 1 baseline.

5.3 User Interface and Insight Requirements

ID	Functional Requirement
FR-4.1	The Streamlit interface shall allow the analyst to select Team 1, Team 2, venue, season, toss winner, and toss decision.
FR-4.2	The interface shall display the predicted winner and win probabilities.
FR-4.3	The interface shall display relevant historical team, venue, toss, and head-to-head insights.
FR-4.4	The system shall provide model-factor/feature-importance information for interpretation.

6. Non-Functional Requirements

ID	Non-Functional Requirement
NFR-1	The prediction interface should return a result within approximately 3 seconds on the project demonstration environment.
NFR-2	The project shall use Git/GitHub for version control.
NFR-3	The application shall provide clear validation messages for invalid or incomplete inputs.
NFR-4	The application shall be usable through a standard desktop web browser during demonstration.
NFR-5	The system shall maintain reproducible preprocessing and model-training steps through documented Python code.
NFR-6	The system shall not require real-time data, user login, payment, or external application access for the core demonstration.

7. Build Plan

7.1 MVP

Requirement	Why in MVP
CSV loading and validation	Required before all downstream processing.
Data cleaning and EDA	Provides reliable project inputs and evidence.
Level 0 and Level 1 baselines	Required benchmarks for model value.
At least one working ML model	Core predictive functionality.
Winner probability output	Core user-facing result.
Streamlit prediction page	Provides a demonstrable application.
Basic historical visualizations	Supports analyst interpretation.

7.2 Build Order

Sprint	Date Range	Deliverable
Sprint 1	05-08-2026 to 20-08-2026	Requirements/SRS and dataset understanding
Sprint 2	21-08-2026 to 05-09-2026	Data cleaning and exploratory data analysis
Sprint 3	06-09-2026 to 21-09-2026	Historical feature engineering and leakage checks
Sprint 4	22-09-2026 to 07-10-2026	Level 0 and Level 1 baseline implementation
Sprint 5	08-10-2026 to 23-10-2026	Logistic Regression, Decision Tree, and Random Forest training
Sprint 6	24-10-2026 to 08-11-2026	Model evaluation, Precision@1, and improvement analysis
Sprint 7	09-11-2026 to 24-11-2026	Streamlit application and prediction workflow
Sprint 8	25-11-2026 to 10-12-2026	Power BI/Tableau dashboard and explainability
Sprint 9	11-12-2026 to 27-12-2026	Testing, documentation, report, and presentation

7.3 Completion Evidence

A sprint is considered complete when the corresponding notebook/code output, test evidence, or application screen is available and documented. Final completion requires a working end-to-end demonstration from user inputs to prediction and supporting insights.

8. Data Requirements

8.1 Main Data Entity: IPL Match

Field / Concept	Use	Constraint
Season	Historical context	Must be available before prediction.
Match date	Chronological ordering and historical feature cutoff	Used to prevent future-data leakage.
Team 1	Prediction input and target definition	Required.
Team 2	Prediction input	Required.
Venue	Historical venue performance	Required where available.
City	Supporting location information	Optional where available.
Toss winner	Pre-match feature	Must refer to the selected match.
Toss decision	Pre-match feature	Must refer to the selected match.
Winner	Target only	Never used as a prediction input.
Post-match result fields	Analysis/validation only	Never used as prediction inputs.

8.2 Data Processing Rules

- Keep the original CSV unchanged as the raw source.
- Create processed data separately.
- Remove or handle duplicates according to documented checks.
- Handle missing values explicitly and document the chosen treatment.
- Standardize inconsistent team and venue names when required.
- Create historical features chronologically so future matches do not influence earlier predictions.
- Use a reproducible train/test evaluation procedure appropriate for chronological sports data.

8.3 Target Definition

For binary classification, the target is defined as 1 when Team 1 wins and 0 otherwise. The model's probability for class 1 represents the estimated probability that Team 1 wins; the probability for Team 2 is derived consistently from the two-class output.

9. Design Decisions and Alternatives Considered

#	Decision	Alternative Considered	Reason
D1	Use one match-level CSV	Use two IPL datasets/files	Simpler, easier to reproduce, and sufficient for the defined pre-match project.
D2	Use pre-match classification	Live/in-match prediction	Matches the available information and keeps the project within scope.
D3	Use Logistic Regression, Decision Tree, Random Forest	Advanced deep learning	These models are appropriate for the taught ML scope and tabular data.
D4	Use Precision@1 as primary metric	Accuracy alone	The system selects one winner; Precision@1 directly evaluates the single selected prediction.
D5	Use chronological historical features	Randomly mix future and past information	Reduces the risk of temporal leakage in sports prediction.

10. Acceptance Criteria

ID	Acceptance Criterion
AC-1	The IPL CSV loads successfully and required columns are validated.
AC-2	Cleaning and preprocessing produce a documented prediction-ready dataset.
AC-3	Level 0 and Level 1 baselines are calculated before final model comparison.
AC-4	At least one trained ML classifier produces a winner prediction and probability.
AC-5	The final model comparison reports Precision@1 for Level 0, Level 1, and each ML model.
AC-6	The Streamlit application accepts the defined pre-match inputs and displays a predicted winner and probabilities.
AC-7	No selected-match winner, final score, margin, player-of-the-match, or other post-match outcome information is used as an input feature.

11. Out of Scope for This Release

- Live or real-time in-match win prediction.
- Betting advice, gambling recommendations, or financial decision support.
- Player-level prediction requiring a separate player-performance dataset.
- Ball-by-ball modelling requiring a second data source.
- Mobile application deployment.
- Advanced deep learning architectures beyond the project's taught scope.
- Use of paid data, login-protected sources, or web scraping for the core dataset.